IDEALS OF A REGULAR LOCAL RING OF DIMENSION 2 WITH REDUCED OR DEPTH-ONE QUOTIENT ARE PRINCIPAL OR TRIVIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the gcd factorization, the height-one and dimension dichotomy, the radical-case arguments, and the depth-one annihilator argument are all retained verbatim.

Editorial note. The label lemma-reduced-quotient-regular-ring-dim-2 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Regular local rings are UFDs. More on Algebra, Lemma more-algebra-lemma-regular-local-UFD. Provides the gcd factorization f_i = f g_i with gcd(g_1, ..., g_r) = 1 underlying the whole argument.
- Dimension theory in a two-dimensional local ring. Standard dimension theory of Noetherian local rings (height-one primes and the maximal ideal as the only point of V(g) in dimension 2). Rules out height-one primes over the g_i, leaving either (g_i) = A or V(g_i) = {m}.

Source reference. Original lemma 0AG0: sources/stacks-chow/tags.txt, source lines 13178–13178. Prerequisite pointer invoked at proof line(s) 4047; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a regular local ring of dimension 2. Let $I \subset \mathfrak{m}$ be an ideal.*

(1) *If A/I is reduced, then $I = (0)$, $I = \mathfrak{m}$, or $I = (f)$ for some nonzero $f \in \mathfrak{m}$.*
(2) *If A/I has depth 1, then $I = (f)$ for some nonzero $f \in \mathfrak{m}$.*

Proof. Assume $I \neq 0$. Write $I = (f_1, \ldots, f_r)$. As A is a UFD (More on Algebra, Lemma 0AG0) we can write $f_i = fg_i$ where f is the gcd of $f_1, \ldots, f_r$. Thus the gcd of $g_1, \ldots, g_r$ is 1 which means that there is no height 1 prime ideal over $g_1, \ldots, g_r$. Then either $(g_1, \ldots, g_r) = A$ which implies $I = (f)$ or if not, then $\dim(A) = 2$ implies that $V(g_1, \ldots, g_r) = \{\mathfrak{m}\}$, i.e., $\mathfrak{m} = \sqrt{(g_1, \ldots, g_r)}$.

Assume A/I reduced, i.e., I radical. If f is a unit, then since I is radical we see that $I = \mathfrak{m}$. If $f \in \mathfrak{m}$, then we see that f^n maps to zero in A/I. Hence $f \in I$ by reducedness and we conclude $I = (f)$.

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Assume A/I has depth 1. Then $\mathfrak{m}$ is not an associated prime of A/I. Since the class of f modulo I is annihilated by $g_1, \ldots, g_r$, this implies that the class of f is zero in A/I. Thus $I = (f)$ as desired. $\square$